LAB 04

Hypothesis Stage: from a Written Card to an Automated Statistical Test

Item	Detail
Topic	Topic 1: Foundations of Quant and Algorithmic Trading: From Thorp to AI Agents
Pipeline stage	Hypothesis
Duration	15 min
Simulator page	http://localhost:8000/#/research/hypothesis
Mode	No coding. Runs ONLINE with your Alpaca paper keys (Integrations page), or OFFLINE on the demo data and the built-in paper broker. Paper trading only.

Lion City Capital and every scenario in these materials are fictitious and for training only. All trading in this course is paper (simulated) trading. Nothing in these materials is investment advice or a recommendation to trade any security.

Scenario

A portfolio manager believes SPY tends to bounce after several losing days. The desk's rule is that no idea enters the signal and backtest stages until it is written as a hypothesis card - before anyone looks at results - and passes an automated statistical test with a decision rule that includes trading costs.

Objective

Write H0 and H1 before looking at results and test them with a Welch t-test and a bootstrap CI.

Before You Start

- Make sure the Trading Simulator is running and open http://localhost:8000 > Course Labs > Lab 04. It is installed once - see the guide 'Download, Install and Start the Trading Simulator Software' in the Learner Guide.
- Record every result you are asked to check - you will use it in the debrief and in the later labs.

Lab Steps - In the Trading Simulator

No coding is needed. Make sure the Trading Simulator is running (installed once - see the guide 'Download, Install and Start the Trading Simulator Software') and open http://localhost:8000. In the sidebar open Course Labs > Lab 04: the app shows the same steps with a Go button that opens each page pre-filled. Every step works offline on the bundled demo data and the built-in paper broker.

Step 1. Write the hypothesis card

Research > Hypothesis test. Keep signal 'losing streak' n = 3, horizon 1 day, alpha 0.05, min edge 3 bps. Read the H1 and H0 text.

Open: http://localhost:8000/#/research/hypothesis (Research > Hypothesis (SPY, losing streak 3))

Check: H0 and H1 are written before you press Run.

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Step 2. Run the test

Press 'Run the test'.

Open: http://localhost:8000/#/research/hypothesis

Check: You can read the signal-day mean, the other-day mean, Welch t and p, and the 95% bootstrap CI.

Step 3. Robustness and multiple testing

Read the robustness grid (other streak lengths and QQQ) and the Bonferroni alpha.

Open: http://localhost:8000/#/research/hypothesis

Check: You can explain why trying many variants inflates false positives.

Expected in the app: A decision (SUPPORTED / NOT SUPPORTED) with p-value, edge in bps and a robustness grid.

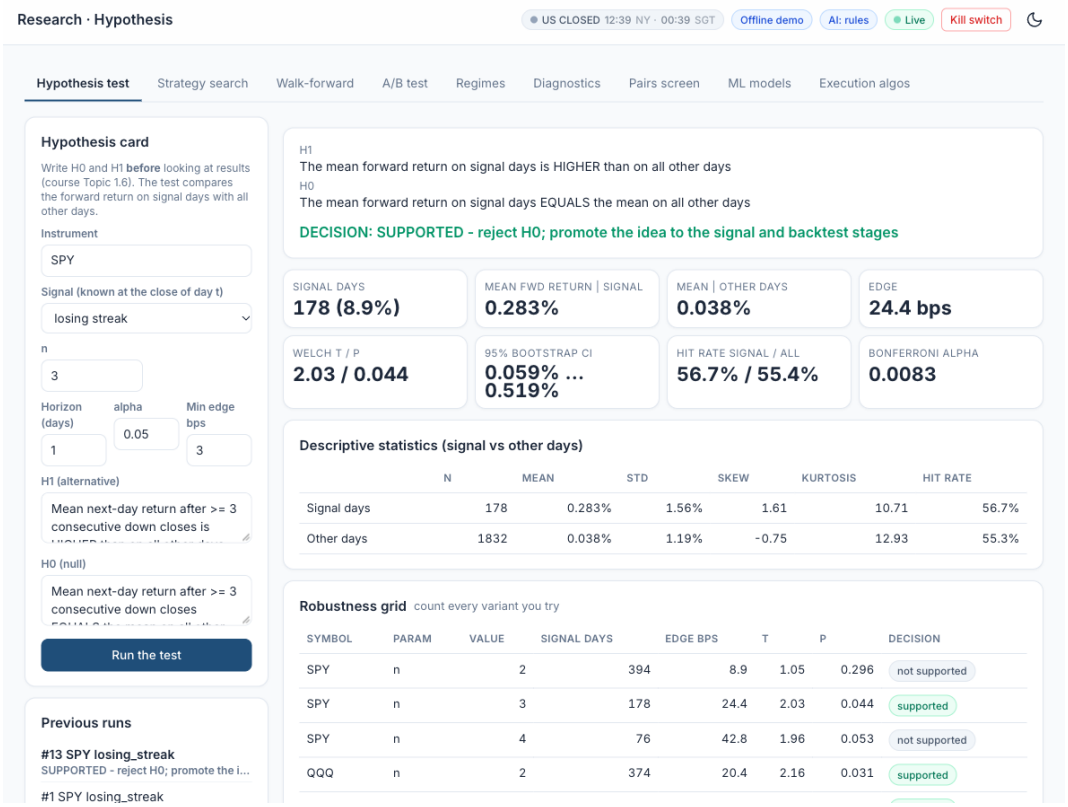


Figure: Lab 04 in the Trading Simulator.

Optional: Let AI Write the Code

This lab needs no coding. To see, run or customise the code behind it, click a link: the AI opens with the prompt filled in and writes the code for you. All prompts are also in labs/Prompt_Library.html and .pdf.

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Test a trading hypothesis - run it in Google Colab
I am a beginner and do not write code. Write ONE complete Python script that runs as-is in a Google Colab notebook. Start with a !pip install line for any library Colab lacks, use pandas, numpy, matplotlib, scipy, scikit-learn, statsmodels and yfinance only, get daily prices from yfinance (auto_adjust=True unless I say otherwise), put a one-line comment above each section, print results as clear tables and charts, and after the code explain in 5 short bullet points how to read the output. Paper trading and education only. Task: test the hypothesis 'after 3 consecutive down days, SPY's next-day return is higher than on other days' on 2018-2025. Write H0 and H1 first, then compare next-day returns after a 3-day losing streak with all other days using a Welch t-test and a bootstrap 95% confidence interval of the difference, and state whether the edge is above 3 basis points at alpha 0.05.

Open in Claude | Open in ChatGPT | Open Google Colab

Debrief

Writing the hypothesis before testing, with a null, a cost-aware decision rule and a count of the variants tried, is the foundation of data-driven trading. AI and machine-learning methods only add value when they test well-posed questions, and larger data sets are how a promising effect in one ETF is confirmed or rejected.

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